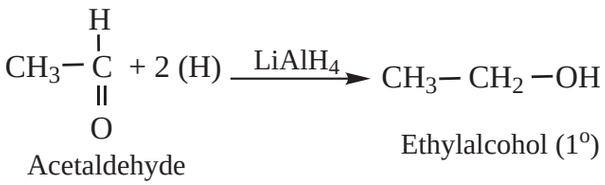
C) Reduction reactions

(i) Reduction to alcohols

We have already learnt that aldehydes and ketones can be easily reduced to primary and secondary alcohols respectively. The most commonly used reducing agents are Lithium Aluminium hydride ($LiAlH_4$), and Sodium borohydride ($NaBH_4$).

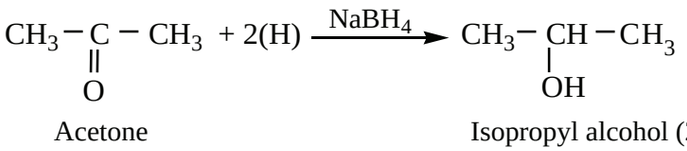
a) Aldehyde are reduced to primary alcohols.

Example



b) Ketone are reduced to Secondary alcohols.

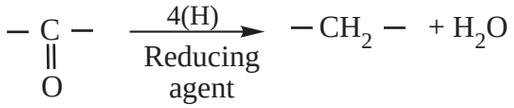
Example



The above reactions can also be carried out with hydrogen in the presence of metal catalyst like Pt, Pd, or Ni. $LiAlH_4$ and $NaBH_4$ do not reduce isolated carbon – carbon double bonds and double bond of benzene rings. In case of α, β unsaturated aldehyde and ketones, $LiAlH_4$ reduces only C = O group leaving C = C bond as such.

ii) Reduction to hydrocarbon

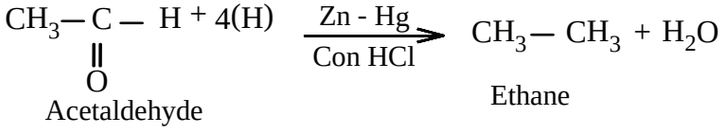
The carbonyl group of aldehydes and ketones can be reduced to methylene group using suitable reducing agents to give hydrocarbons.

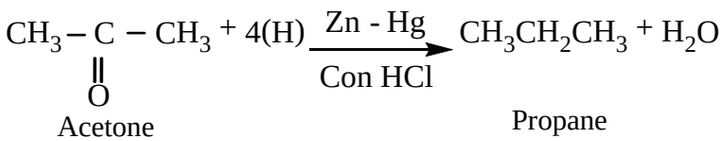


a) Clemmensen reduction

Aldehydes and Ketones when heated with zinc amalgam and concentrated hydrochloric acid gives hydrocarbons.

Example





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